

Beyond Morin, and deeper into A_7 :

the corank filtration of Thom polynomials, and the anatomy of Chern positivity at $d \leq 7$

Chern++

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Abstract

We place Rimányi’s conjecture inside a filtration $C_1 \subset C_2 \subset \dots$ of cones of symmetric functions, running from the Chern-monomial cone to the Schur-positive cone. Rimányi’s conjecture says that $\text{Tp}_{A_d} \in C_1$. The natural generalisation is $\text{Tp}_\eta \in C_{\text{corank } \eta}$. We prove the matching lower bound, and we prove, by exact certificates, that the generalisation itself is false. The minimal rank $\rho(\eta)$ is therefore a new invariant, and we tabulate it on the published registry. For Morin singularities we give a sweep algorithm for the chamber series that is 50–100× faster than the previous one. It yields Tp_{A_7} exactly for $\ell \leq 5$ and Tp_{A_6} for $\ell \leq 8$, all positive. We locate where positivity is tight (the plane sector, whose values are Kreweras numbers), and we observe a quantitative strengthening: the negative mass of every packet is at most a fixed fraction of a single term. Every statement is labelled *proved*, *verified* (a finite exact computation) or *conjectured*.

Summary of results

	statement	status
2	$\rho(\eta) \geq \text{support corank} \geq \text{corank } \eta$	proved
3	$\sum \text{Chern coefficients} = [s_{(n)}] \text{Tp}; = 0 \text{ at corank} \geq 2$	proved
4	$\text{Tp}_{I_{2,4}}, \text{Tp}_{III_{2,3}} \notin C_2$; $\rho \neq \text{corank}$	proved
Table 1	ρ on the registry; $\rho - \text{corank}$ reaches 2	verified
§2	Tp_{A_7} ($\ell \leq 5$), Tp_{A_6} ($\ell \leq 8$) exact and positive, $\min C = 1$	verified
7	plane sector: $C(M) = \text{Kreweras number}$, $d \leq 7$	verified
8	$N(M) \leq \kappa_d A_{\beta_{\max}(M)}$ with $\kappa_5 = \kappa_6 = \frac{1}{6}$, $\kappa_7 = \frac{8}{21}$	conjectured
§4	0-Hecke landscape P_5, P_6 (stable), P_7 (on boxes)	verified

1 The corank filtration

Write a Thom polynomial in the relative Chern classes and read c_k as the complete homogeneous symmetric function h_k . Then $s_\nu = \det(c_{\nu_i + j - i})$, and Pragacz–Weber positivity [3] says that every Tp_η is a nonnegative combination of the s_ν .

Definition 1. For $r \geq 1$ let C_r be the convex cone spanned by all products $s_{\nu^1} \cdots s_{\nu^k}$ in which every ν^i has at most r rows. Set $\rho(\eta) = \min\{r : \text{Tp}_\eta \in C_r\}$.

Since $s_{(k)} = c_k$, the cone C_1 is spanned by the Chern monomials; because Chern monomials form a basis, $f \in C_1$ exactly when all its Chern coefficients are nonnegative. At the other end, $\bigcup_r C_r$ is the Schur-positive cone. Hence:

$$\text{Rimányi's conjecture} \iff \rho(A_d) = 1, \quad \text{Pragacz–Weber} \iff \rho(\eta) < \infty.$$

The filtration is natural from the point of view of residue formulas. The Bérczi–Szenes formula is a residue over $\mathbb{P}(TX)$: after the residue, each chamber monomial becomes a product of one-row Schur functions $c_{\ell+1+\alpha_i}$, and Rimányi’s conjecture asks the packet sums to be nonnegative. A formula of the same kind on the Grassmannian $\text{Gr}(r, TX)$ would produce products of Schur functions with at most r rows. So a *corank- r residue formula with nonnegative packet sums would place Tp_η in C_r .*

Theorem 2 (lower bound; proved). *If every Schur component of Tp_η has at least r rows, then $\text{Tp}_\eta \notin C_{r-1}$. In particular $\rho(\eta) \geq \text{corank } \eta$.*

Proof. Let $g = s_{\nu^1} \cdots s_{\nu^k}$ with every $\ell(\nu^i) \leq r-1$. By the Littlewood–Richardson rule, $s_{\nu^1+\dots+\nu^k}$ (row-wise sum) appears in g with coefficient 1, and it has at most $r-1$ rows. All LR coefficients are nonnegative, so in a nonnegative combination of such products nothing cancels it. Every nonzero element of C_{r-1} therefore has a Schur component with at most $r-1$ rows, and Tp_η has none. A singularity of corank r lies in the Thom–Boardman stratum Σ^r , and every Schur component of its Thom polynomial contains the rectangle $(\ell+r)^r$. This is the support property, verified on every registry table: the column s of Table 1 is at least the corank. $\square$

Corollary 3 (proved). *The sum of the Chern coefficients of any Tp_η equals its coefficient on $s_{(n)}$, where $n = \text{codim } \eta$. This is $(d!)^{\ell+1}$ for A_d and 0 for every η of corank ≥ 2 . In particular, no Thom polynomial of corank ≥ 2 is Chern-positive, and each has at least one negative Chern coefficient.*

Proof. $h_\mu = \sum_\lambda K_{\lambda\mu} s_\lambda$, and $K_{(n),\mu} = 1$ for every $\mu \vdash n$. So $[s_{(n)}] \sum_\mu a_\mu h_\mu = \sum_\mu a_\mu$. At corank ≥ 2 every component has at least two rows. For A_d the value $(d!)^{\ell+1}$ is verified: on every published table (tier 14) and on all fifteen new tables of §2. $\square$

The negative Chern coefficients of non-Morin Thom polynomials are thus *forced*. They are not an accident of the data, and they are exactly what Theorem 2 forbids. The natural generalisation of Rimányi’s conjecture would be $\rho = \text{corank}$. It is false.

Proposition 4 (proved by certificate). $\text{Tp}_{I_{2,4}} = 16s_{42} + 4s_{33} + 12s_{321} + 5s_{222} + 2s_{2211}$ (at $\ell = 0$) is not in C_2 . Similarly $\text{Tp}_{\text{III}_{2,3}} = 8s_{53} + 4s_{431} + 2s_{332}$ (at $\ell = 1$) is not in C_2 . Both lie in C_3 .

Proof. Since products of Schur functions are Schur-positive, a decomposition can only use generators g with $\text{supp } g \subseteq \text{supp } \text{Tp}$, and there are finitely many. For $I_{2,4}$ the functional $\varphi = [s_{33}] + [s_{222}] - [s_{321}]$ is nonnegative on each of them but $\varphi(\text{Tp}) = 4 + 5 - 12 = -3$. For $\text{III}_{2,3}$ take $\varphi = [s_{332}] - [s_{431}]$: for instance $s_{33}s_2 = s_{53} + s_{431} + s_{332}$, while $\varphi(\text{Tp}) = -2$. The nonnegativity checks, and the C_3 decompositions, are exact rational computations in `chernpp.corank` (tier 14). $\square$

The invariant ρ on the registry. Table 1 gives ρ for every surveyed table. Each entry is decided by a linear programme whose answer is re-verified in exact arithmetic, with a decomposition or a Farkas vector as appropriate. The Farkas vectors are strikingly simple: every one found is $\pm \frac{1}{k}$ on a handful of Schur functions. What the table shows:

- $\rho(I_{a,b}) = 2$ exactly when $b \leq a+1$, and 3 otherwise, for $a \leq b$. Likewise $\rho(\text{III}_{a,b}) = 2$ exactly when $a = b$. In both families ρ is constant in ℓ wherever it was computed.
- ρ is not always constant in ℓ : $\rho(B_4) = 2$ at $\ell = 1$ and 3 at $\ell = 2$.
- $\rho - s$ reaches 2: $B_{5,3}$ at $\ell = 1$ has $s = 2$ and $\rho = 4$. C_6 and $C_{5,3}$ have $s = 3$ and $\rho = 4$.

We have not found an invariant of the local algebra that predicts ρ in general; the multiplicity and the Hilbert function do not. This is the open question the filtration raises.

Remark 5 (a negative result). A sharper variant replaces cones by a basis: $b_\lambda^{(r)} = \prod_k s_{(\lambda_{rk+1}, \dots, \lambda_{rk+r})}$. It is unitriangular against h_λ , equals h_λ at $r = 1$, and so has unique coefficients, as Rimányi's conjecture does. Tp is not $b^{(\text{corank})}$ -positive in general. $I_{2,2}$, $I_{3,3}$, $I_{4,4}$, $\text{III}_{2,2}$ and $\text{III}_{3,3}$ are $b^{(2)}$ -positive wherever tabulated, and $I_{3,4}$ at $\ell = 0$. But $\text{III}_{4,4}$ is not, $I_{2,3}$ fails from $\ell = 2$, and $I_{2,4}$, $I_{4,5}$ and B_4 fail already at the smallest ℓ .

Table 1: $\rho(\text{Tp}) = \min\{r : \text{Tp} \in C_r\}$ on the non-Morin tables of the registry, by relative dimension ℓ , with the support corank s (every Schur component contains $(\ell + s)^s$). Every entry is decided with an exact certificate; a dash means not in the registry or not yet surveyed.

singularity	s	$\ell = 0$	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$	$\ell = 5$
$I_{2,2}$	2	2	2	2	2	2	—
$I_{2,3}$	2	2	2	2	2	—	—
$I_{2,4}$	2	3	3	3	—	—	—
$I_{2,5}$	2	3	3	—	—	—	—
$I_{2,6}$	2	3	—	—	—	—	—
$I_{2,7}$	2	3	—	—	—	—	—
$I_{3,3}$	2	2	2	2	—	—	—
$I_{3,4}$	2	2	2	—	—	—	—
$I_{3,5}$	2	3	—	—	—	—	—
$I_{3,6}$	2	3	—	—	—	—	—
$I_{4,4}$	2	2	—	—	—	—	—
$I_{4,5}$	2	2	—	—	—	—	—
$\text{III}_{2,2}$	2	—	2	2	2	2	2
$\text{III}_{2,3}$	2	—	3	3	3	3	—
$\text{III}_{2,4}$	2	—	3	3	3	—	—
$\text{III}_{2,5}$	2	—	3	3	—	—	—
$\text{III}_{2,6}$	2	—	3	—	—	—	—
$\text{III}_{3,3}$	2	—	2	2	2	—	—
$\text{III}_{3,4}$	2	—	3	3	—	—	—
$\text{III}_{3,5}$	2	—	3	—	—	—	—
$\text{III}_{4,4}$	2	—	2	—	—	—	—
$(x^2 + y^3, y^4)$	2	2	—	—	—	—	—
$(x^2 + y^4, xy^2)$	2	3	—	—	—	—	—
B_4	2	—	2	3	—	—	—
$B_{5,1}$	2	2	2	2	—	—	—
$B_{5,2}$	2	—	2	2	—	—	—
$B_{5,3}$	2	—	4	—	—	—	—
B_6	2	2	2	—	—	—	—
C_3	3	—	—	—	3	—	—
$C_{4,1}$	3	—	—	3	—	—	—
$C_{4,2}$	3	—	—	3	—	—	—
$C_{5,1}$	3	—	3	3	—	—	—
$C_{5,2}$	3	—	3	3	—	—	—
$C_{5,3}$	3	—	—	4	—	—	—
$C_{5,4}$	3	—	3	—	—	—	—
C_6	3	—	4	—	—	—	—
Σ^3 open (cup)	3	3	—	—	—	—	—

2 New exact data for A_6 and A_7

The sweep. Every monomial of a level- ℓ denominator factor of $F_d = \mathcal{N}_d / \prod_r (1 - f_r)$ contains $x_{\ell-1}$ exactly once. Hence $H = G + fH$ is a recurrence along one axis, and multiplying by $1/(1 - f_r)$ needs one pass over the grid instead of a fixed point. At $d = 7$ this is 34 passes, not ~ 4500 . Values are computed modulo word-sized primes, reconstructed by CRT, and checked against a prime held back from the reconstruction (`chernpp.bboxes`). The sweep agrees cell by cell with the fixed point of `chernpp.chern`, and packet by packet with `chernpp.crt`.

Boxes and packets. Every ballot ordering of M satisfies $\beta \leq \beta_{\max}(M)$, the partial sums of the decreasing ordering. So the level box $\beta_j \leq (d - j)L$ holds every packet with $\min M \geq -L$, and the charge box $\beta_j \leq jp$ holds every packet with $\max M \leq p$. Since $C(M)$ does not depend on ℓ , one level box gives Tp_{A_d} at every $\ell < L$.

Observation 6 (verified). Tp_{A_7} for $\ell \leq 5$ (8033 Chern coefficients) and Tp_{A_6} for $\ell \leq 8$ (12692) have every coefficient positive, and the minimum is 1. Both agree with every published table they overlap. The A_7 box has 49.6M cells and took seven minutes with four primes. The data are in `results/morin_a{6,7}.json`.

3 Where positivity is tight

Observation 7 (verified, $d \leq 7$). On the plane sector $\max M \leq 1$, write $M = (1^{d-b}, 1 - s_1, \dots, 1 - s_b)$ with $s_i \geq 1$. Then

$$C(M) = \frac{d!}{(d - b + 1)! \prod_i m_i!},$$

the number of noncrossing partitions of $[d]$ with block sizes s_i (Kreweras), where m_i is the number of blocks of size i . This holds for every plane-sector packet at $d \leq 7$. The external summary [4] proves positivity on this sector; the observation identifies the values. In particular $C(M) = 1$ on the sector exactly at $M = 0$ and $M = (1^{d-1}, 1 - d)$, and these are the global minima.

Off the plane sector the margin is large. The smallest $C(M)$ with $\max M = 2$ is 12, 17, 27 at $d = 5, 6, 7$, and the layer minima grow quickly with the charge (Table 2).

Anatomy of a packet. Let $N(M)$ be the negative mass of a packet, $\sum_{A_\beta < 0} |A_\beta|$ over its ballot orderings, and $A_{\text{dom}}(M) = A_{\beta_{\max}(M)}$ the coefficient of its decreasing ordering. Table 2 gives, per charge layer, the largest ratio N/A_{dom} .

Conjecture 8 (dominance). For every d there is $\kappa_d < 1$ with $N(M) \leq \kappa_d A_{\beta_{\max}(M)}$ for every zero-sum M . In particular $C(M) \geq (1 - \kappa_d) A_{\beta_{\max}(M)} > 0$.

The evidence: $\kappa_5 = \kappa_6 = \frac{1}{6}$ and $\kappa_7 = \frac{8}{21}$ on every packet with $\max M \leq 8, 6, 5$ respectively (exact). The extremal packets are $(1, 1, 0, -1, -1)$, $(1, 1, 1, 0, -1, -2)$ and $(2, 1, 1, 0, -1, -1, -2)$. In every layer the ratio falls by a factor of 3 to 4 per unit of charge, so the conjecture is a statement about small charge. Three remarks keep it honest.

- Individual coefficients are unbounded below: $\min A_\beta = -3622$ in the $d = 7$ layer $p = 5$. This corrects the report's earlier "least value -2 ".
- κ_d grows with d . The conjecture is informative only if it stays below 1, and $d = 8$ would test that.
- The single term $A_{\beta_{\max}(M)}$ is the largest in all but 42 of the 3538 packets at $d = 7$. A proof would need a lower bound for this one coefficient of F_d and an upper bound for the negative ones, and no pairing is involved. This is the kind of statement the report's Assessment §4 asked for.

Table 2: Charge layers max $M = p$ (every packet complete). $N(M)$ is the negative mass of the packet and A_{dom} the coefficient of its decreasing ordering $\beta_{\text{max}}(M)$.

d	p	packets	with $A_\beta < 0$	$\min C(M)$	$\min A_\beta$	$\max N(M)/A_{\text{dom}}$
5	1	6	1	1	-1	1/6
5	2	23	3	12	-1	1/20
5	3	54	3	38	-2	1/80
5	4	108	3	24	-3	1/258
5	5	185	3	336	-4	$1.1e - 03$
5	6	297	3	1296	-5	$2.8e - 04$
5	7	441	5	3696	-6	$6.8e - 05$
5	8	632	5	2880	-10	$1.6e - 05$
6	1	10	2	1	-1	1/6
6	2	47	7	17	-3	1/12
6	3	141	18	55	-6	3/103
6	4	333	35	202	-20	3/296
6	5	674	52	120	-70	$3.5e - 03$
6	6	1226	73	2400	-252	$1.2e - 03$
7	1	14	5	1	-4	3/8
7	2	90	43	27	-26	8/21
7	3	331	134	154	-120	43/350
7	4	931	339	326	-516	9/223
7	5	2172	659	1284	-3622	$1.6e - 02$

4 The 0-Hecke landscape

On a packet, let π_i act by $(\pi_i f)(\alpha) = f(\alpha) + f(s_i \alpha)$ if $\alpha_i > \alpha_{i+1}$, by $f(\alpha)$ if $\alpha_i = \alpha_{i+1}$, and by 0 otherwise. The π_i satisfy the 0-Hecke relations (tier 11), and Π_{w_0} puts $C(M)$ on the decreasing ordering. The operators and this criterion are from [4, §3.3]. So $P_d = \{w : \Pi_w F_d \geq 0\}$ contains w_0 iff Rimányi's conjecture holds. P_d is an upper set in the left weak order, and its minimal elements are the minimal cancellation units.

Table 3: The 0-Hecke positivity set P_d restricted to the packets complete in the level box $\min M \geq -L$.

d	L	packets	with $A_\beta < 0$	$ P_d $	minimal elements
5	5	377	12	114	4
6	4	532	39	643	11
6	5	1206	76	643	11
7	2	105	39	3998	23
7	3	436	141	3352	49
7	4	1367	374	2792	30
7	5	3539	845	2696	33

- $P_5 = S_5 \setminus \langle s_3, s_4 \rangle$, with minimal elements $s_1, s_2, s_2 s_3, s_2 s_3 s_4$. This is unchanged from $L = 4$ to 5.
- $|P_6| = 643$ with 11 minimal elements, unchanged from $L = 4$ to 5. $s_1 \in P_6$, which is the unpaired-tail and paired inequalities of the τ frame holding at $d = 6$.
- At $d = 7$, $s_1 \notin P_7$, which is the failure of the τ frame at $d = 7$. P_7 still shrinks with the box ($|P_7| = 3998, 3352, 2792, 2696$ for $L = 2, \dots, 5$). The minimal parabolic subsets J with

$w_0(J) \in P_7$ are the same at $L = 4$ and 5 : $\{1, 2, k\}$ ($k = 3, \dots, 6$) and $\{1, 3, 5\}$. At $d = 5$ they are $\{1\}$ and $\{2\}$, and at $d = 6$ $\{1\}$ and $\{2, 5\}$. The smallest unit of symmetrisation that clears every negative grows from one transposition to three, so no fixed-size pairing survives.

Exclusions from P_d are proofs, each witnessed by a packet. Memberships hold on the stated boxes only.

5 Stabilisation, and what this means for $\mathcal{Q}_8$

Observation 9 (verified, $d = 5, 6, 7$). $F_d|_{x_{d-1}=0} = F_{d-1}$, cell by cell, on the tested boxes. The corresponding statement for numerators, $[z_d^{e(d)}]\mathcal{Q}_d = (-1)^{e(d)}\mathcal{Q}_{d-1}$ with $e(d) = \lfloor d^2/4 \rfloor - d + 1$, was checked directly at $d = 5$.

The corank filtration changes the case for $\mathcal{Q}_8$. Evidence about generalisations of the conjecture now comes from breadth across coranks, which the registry already supplies. Depth in d adds one Morin data point, and the published Tp_{A_8} and Tp_{A_9} at $\ell = 0$ already lie in C_1 . $\mathcal{Q}_8$ matters for one question raised here: whether $\kappa_8 < 1$ in Conjecture 8. The constraints already available for any candidate $\mathcal{Q}_8$ are its degree (22), its normalisation, the stabilisation $[z_8^9]\mathcal{Q}_8 = -\mathcal{Q}_7$, and agreement with Rimányi’s Tp_{A_8} at $\ell = 0$.

Reproducing

- `tools/morin_tables.py 7 6 results/morin_a7.json` (the A_7 tables).
- `tools/packet_anatomy.py`, `tools/hecke_landscape.py`, and `tools/corank_survey.py` for ρ .
- `tools/render_d7_tables.py report/tables` regenerates every table here.

Tiers 10–14 of the test suite pin each module.

References

- [1] L. Bérczi, A. Szenes, Thom polynomials of Morin singularities, *Ann. of Math.* 175 (2012) 567–629.
- [2] R. Rimányi, Thom polynomials, symmetries and incidences of singularities, *Invent. Math.* 143 (2001) 499–521.
- [3] P. Pragacz, A. Weber, Positivity of Schur function expansions of Thom polynomials, *Fund. Math.* 195 (2007) 85–100.
- [4] Rimányi positivity: findings summary (29 July 2026), unrefereed, `papers/rimanyi_positivity_findings_summary.pdf`.